

Enthalpy and Calorimetry

Answer key with worked solutions

A. Understanding

1. Classify.

(a) **exothermic**, ΔH negative (b) **endothermic**, ΔH positive (heat is absorbed to melt the ice) (c) **endothermic**, ΔH positive (the solution cools because heat is absorbed) (d) **exothermic**, ΔH negative

2. Thermochemical equations.

(a) $2 \times (-890.5 \text{ kJ}) = -1781.0 \text{ kJ}$ (b) $+890.5 \text{ kJ}$ (c) $n = \frac{8.02 \text{ g}}{16.04 \text{ g/mol}} = 0.500 \text{ mol}$; heat released $= 0.500 \text{ mol} \times 890.5 \text{ kJ/mol} = 445 \text{ kJ}$ ($\Delta H = -445 \text{ kJ}$)

B. $q = mc\Delta T$

3. Copper.

$$q = (40.0 \text{ g})(0.385 \text{ J/(g} \cdot ^\circ\text{C)})(75.0 ^\circ\text{C} - 25.0 ^\circ\text{C}) = 770. \text{ J (g and } ^\circ\text{C cancel)}$$

4. Specific heat.

$$c = \frac{q}{m\Delta T} = \frac{900. \text{ J}}{(100. \text{ g})(20.0 ^\circ\text{C})} = 0.450 \text{ J/(g} \cdot ^\circ\text{C)} \text{ (close to iron, } 0.449 \text{ J/(g} \cdot ^\circ\text{C))}$$

5. Final temperature.

$$\Delta T = \frac{q}{mc} = \frac{5.00 \times 10^3 \text{ J}}{(150. \text{ g})(4.184 \text{ J/(g} \cdot ^\circ\text{C)})} = 7.97 ^\circ\text{C}$$

$$T_{\text{final}} = 22.0 ^\circ\text{C} + 7.97 ^\circ\text{C} = 30.0 ^\circ\text{C}$$

C. Calorimetry

6. Ammonium nitrate.

$$\Delta T = 16.0\text{ }^{\circ}\text{C} - 22.0\text{ }^{\circ}\text{C} = -6.0\text{ }^{\circ}\text{C}$$

$$q_{\text{soln}} = (108.0\text{ g})(4.184\text{ J/(g}\cdot^{\circ}\text{C)})(-6.0\text{ }^{\circ}\text{C}) = -2711\text{ J, so } q_{\text{reaction}} = +2711\text{ J}$$

$$n = \frac{8.00\text{ g}}{80.04\text{ g/mol}} = 0.09995\text{ mol}$$

$$\Delta H = \frac{+2711\text{ J}}{0.09995\text{ mol}} = +2.7 \times 10^4\text{ J/mol} = \mathbf{+27\text{ kJ/mol}}$$
 (endothermic; 2 significant figures from ΔT)

7. Neutralization.

$$\text{Mass of solution} = 50.0\text{ mL} \times 1.00\text{ g/mL} = 50.0\text{ g}; \Delta T = 33.5\text{ }^{\circ}\text{C} - 20.0\text{ }^{\circ}\text{C} = 13.5\text{ }^{\circ}\text{C}$$

$$q_{\text{soln}} = (50.0\text{ g})(4.184\text{ J/(g}\cdot^{\circ}\text{C)})(13.5\text{ }^{\circ}\text{C}) = 2824\text{ J, so } q_{\text{reaction}} = -2824\text{ J}$$

$$n(\text{H}_2\text{O}) = 2.00\text{ mol/L} \times 0.0250\text{ L} = 0.0500\text{ mol}$$

$$\Delta H = \frac{-2824\text{ J}}{0.0500\text{ mol}} = -5.65 \times 10^4\text{ J/mol} = \mathbf{-56.5\text{ kJ/mol}}$$

D. Challenge

8. Ethanol under a can.

$$\text{(a) Heat released: } \frac{0.500\text{ g}}{46.07\text{ g/mol}} \times 1366.8\text{ kJ/mol} = 14.83\text{ kJ}$$

$$\Delta T_{\text{max}} = \frac{1.483 \times 10^4\text{ J}}{(200.\text{ g})(4.184\text{ J/(g}\cdot^{\circ}\text{C)})} = 17.7\text{ }^{\circ}\text{C}$$

$$\text{(b) Percentage transferred} = \frac{12.4\text{ }^{\circ}\text{C}}{17.73\text{ }^{\circ}\text{C}} \times 100\% = \mathbf{69.9\%}$$
 (the rest heated the can and the air)

9. Iron in water.

Heat lost by iron = heat gained by water:

$$(50.0\text{ g})(0.449\text{ J/(g}\cdot^{\circ}\text{C)})(100.0\text{ }^{\circ}\text{C} - T) = (100.\text{ g})(4.184\text{ J/(g}\cdot^{\circ}\text{C)})(T - 20.0\text{ }^{\circ}\text{C})$$

$$22.45\text{ J/}^{\circ}\text{C} \times (100.0\text{ }^{\circ}\text{C} - T) = 418.4\text{ J/}^{\circ}\text{C} \times (T - 20.0\text{ }^{\circ}\text{C})$$

$$2245\text{ J} + 8368\text{ J} = (418.4 + 22.45)\text{ J/}^{\circ}\text{C} \times T, \text{ so } T = \frac{10613\text{ J}}{440.85\text{ J/}^{\circ}\text{C}} = \mathbf{24.1\text{ }^{\circ}\text{C}}$$